EE2080: Lecture 04: 10 Aug 2026

RECAP:

1) IID Process: $\{X_t : t \in \mathcal{T}\}$ IID process with common CDF F iff $\forall n \in \mathbb{N}$ &
$\forall t_1, \dots, t_n \in \mathcal{T}$, $F_{X_{t_1}, \dots X_{t_n}}(x_1, \dots, x_n) = \prod_{i=1}^{n} F(x_i)$, $x_i \in \mathbb{R}$.

2) Gambler's Ruin: Initial wealth: k : Target wealth: N_0.
$X_0 = k, \{X_i\}_{i \in \mathbb{N}} \overset{iid}{\sim} Ber(p)$. Success prob: $S_k = \dfrac{1-(q/p)^k}{1-(q/p)^{N_0}}$. $q = 1-p$, $p \neq 1/2$
$p = 1/2$ $S_k = k/N_0$.

3) Bernoulli Process: $\{Z_i\}_{i \in \mathbb{N}} \overset{iid}{\sim} Ber(p)$ represents arrivals/no arrivals at i.
$Z_i = 1$ $Z_i = 0$
Interarrival times: $\{X_i\}_{i \in \mathbb{N}} \overset{iid}{\sim} Geo(p)$.
Aggregate # arrivals upto instant i $\{Y_i\}_{i \in \mathbb{N}} \sim Bin(i, p)$.

4) Arrival Process: A seq. of increasing RVs $0 < S_1 < S_2 < \dots$ representing arrival epochs $(S_{i+1} - S_i > 0)$.
Equivalent descriptions: i) Inter-arrival times $\{X_i\}_{i \in \mathbb{N}}$.
ii) Counting Process: $\{N_t\}_{t \geq 0}$ with $N_0 = 0$ with p. 1.

5) Renewal Process: Arrival process in which the inter-arrival times $X_1, X_2, \dots$ is a sequence of positive IID RVs.

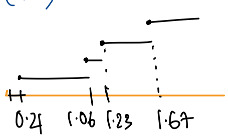
6) Poisson Process: A renewal process in which the inter-arrival times have an exponential distribution.

Example: Let the arrival rate be $\lambda = 2$ arrivals per minute

Description 1: Using inter-arrival times.

$X_1, X_2, \dots \overset{iid}{\sim} \exp(\lambda)$

	X_i (min)	Arrival S_i (min)
1)	0.21	0.21
2)	0.85	1.06
3)	0.17	1.23
4)	0.44	1.67



t	0	0.5	1	1.5	2
N_t	0	1	1	3	4

Description 2: Using Poisson Counts:

Instead of generating arrival times, we find the prob of seeing a certain # of arrivals at a given time:

$X \sim Poi(\lambda) \Rightarrow$
$P_X(k) = \dfrac{\lambda^k e^{-\lambda}}{k!}$

$N_t \sim Poisson(\lambda t)$
$N_2 \sim Poisson(4)$ $\mathbb{E}[N_2] = 4$

PLAN: Poisson Process Continued ...

Exponential Distribution and Memoryless Property

Consider a random variable X defined on $(\Omega, \mathcal{F}, \mathbb{P})$ s.t $X \sim \exp(\lambda)$.

- PDF: $f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$
- CDF: $F_X(x) = 1 - e^{-\lambda x}$, $x \in \mathbb{R}$
- Complementary CDF: $\mathbb{P}(X > x) = e^{-\lambda x}$, $x \in \mathbb{R}$
- Memoryless Property: $\mathbb{P}(X > s+t \mid X > s) = \mathbb{P}(X > t)$, $s, t \in \mathbb{R}$

  Prob. of waiting for an additional time of t given that the waiting time has already exceeded s is the same as the prob. of waiting for t from the beginning

  $$\mathbb{P}(X > s+t) = \mathbb{P}(X > s)\,\mathbb{P}(X > t).$$

Qn: $X \sim \exp \Rightarrow X$ memoryless.

- For a continuous RV X, does the reverse implication hold?

  i.e., is exponential distribution the only continuous distribution that has the memoryless property?

Qn: Are regular arrivals memoryless?

For example: a bus arrives at a bus stop A once every 15 minutes.

Qn: If the memoryless property is restricted to integer values i.e., if

$\mathbb{P}(X > t+k \mid X > t) = \mathbb{P}(X > k)$ should hold for only for $k, t \in \mathbb{Z}^+$,

then which distribution has memoryless property?

Bernoulli process which has IID geometric inter-arrival times is like a discrete-time version of the Poisson process (which has IID exponential inter-arrival times)

Example: The M/M/1 queue

service process · number of servers · arrival process

M: memoryless
D: Deterministic
G: General

The first M $\Rightarrow$ arrival process is memoryless $\Rightarrow$ Poisson process (IID exp. distributed inter-arrival times)

The second M $\Rightarrow$ service process is memoryless $\Rightarrow$ IID exp. distributed service times.

1 $\rightarrow$ single server.

M/M/1 queue $\Rightarrow$ A queueing system with a Poisson arrival ($\exp(\lambda)$) and a single server that serves arriving customers in FCFS order with service times $\sim \exp(\mu)$. The service times are independent of each other and of the inter-arrival times.

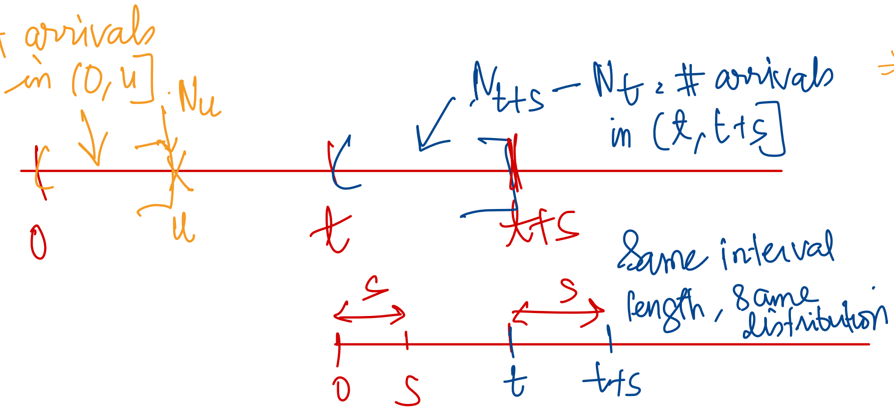
Defn: A point process $\{N_t\}_{t\geq 0}$ is a random process taking values in $0,1,2,\dots$, such that $N(0)=0$, and for each ω, $N(t,\omega)$ is a non-decreasing, right continuous step function.
A point process is essentially characterised by a random distribution of points on $\mathbb{R}^+$ (i.e., the jump instants) and a sequence of integer-valued random variables corresponding to the jump at each point. If we think of a batch of arrivals, then N_t is the number of arrivals in the interval $[0,t]$.

Defn: (Poisson Process): A point process $\{N_t\}_{t\geq 0}$ is a Poisson process if

a) all jumps of N_t are of unit size

b) $\forall\, t,s \geq 0$ $\quad N_{t+s}-N_t \perp\!\!\!\perp N_u \quad \forall\, u\leq t \longrightarrow$ INDEPENDENT INCREMENTS.

c) $\forall\, t,s\geq 0$ the distribution of $N_{t+s}-N_t$ does not depend on t – STATIONARY INCREMENTS.



$N_u \perp\!\!\!\perp N_{t+s}-N_t$
$\Rightarrow$ # arrivals in $(0,u] \perp\!\!\!\perp$ # arrivals in $(t,t+s]$ $\quad u\leq t$
Non-overlapping intervals

N_s identically distributed as $N_{t+s}-N_t$.

Example related to Poisson Process that is not a Renewal Process

Cox PROCESS : (also called a doubly stochastic Poisson Process).

Step 1: Choose a random arrival rate

$$\Lambda = \begin{cases} 1 \text{ arrival/minute with prob } 1/2 \\ 3 \text{ arrivals/minute with prob } 1/2 \end{cases}$$

Step 2 : Once $\Lambda = \lambda$ is fixed inter arrival times are $X_i \overset{iid}{\sim} \exp(\lambda)$.

Example: Suppose $\Lambda = 1$

Arrival #	inter-arrival X_i	Arrival epoch S_i
1	0.72	0.72
2	1.53	2.25
3	0.91	3.16
4	0.44	3.60

Mean inter-arrival time is 1.

$\Lambda = 3$

inter-arrival X_i	Arrival epoch S_i
0.21	0.21
0.08	0.29
0.43	0.72
0.19	0.91

Arrivals are denser because the mean inter-arrival time is $1/3$.

The observer only sees 0.72, 1.53, 0.91, 0.44, etc. (OR) 0.21, 0.08, 0.43, 0.19, ...

Distribution of inter-arrival time $f_X(x) = \frac{1}{2} e^{-x} + \frac{3}{2} e^{-3x}$. $x \geq 0$.

$\therefore$ Inter arrival times are identically distributed.

But are they independent?

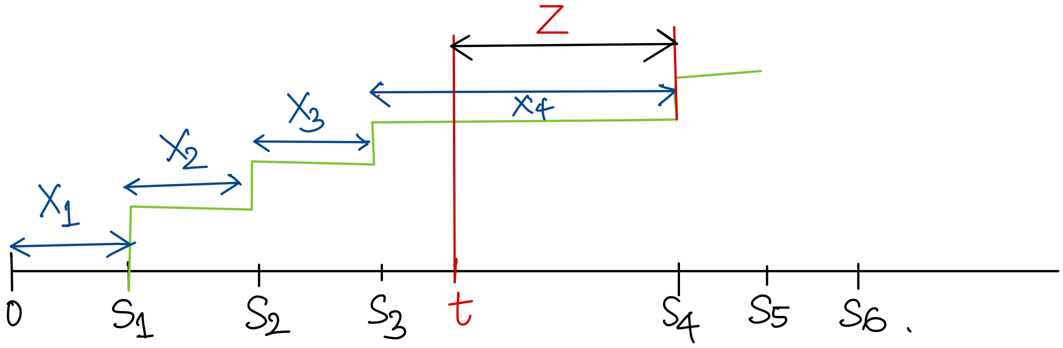
Intuition: Think of waiting for buses: There are 2 kinds of possible days:

i) Heavy traffic days: buses come every 5 minutes on average.

ii) Light traffic days: buses come every 15 minutes on average.

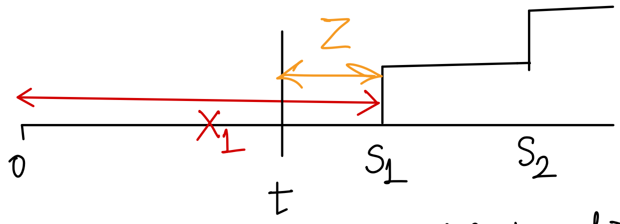
If you wait for the first bus for a long time (say > 10 minutes), you naturally conclude that today is probably a light traffic day. Consequently, you expect the next bus to take longer to come. Thus observing the first waiting time gives info about the second waiting time.

Theorem: For a Poisson process of rate λ, and any given $t > 0$, the length of the interval from t until the first arrival after t is a positive RV Z with the CDF $F_Z(z) = 1 - \exp(-\lambda z)$, $z \geq 0$. This RV is independent of both N_t and the N_t arrival epochs $S_1, S_2, \ldots, S_{N_t}$ before time t.



Implication: The portion of a Poisson process starting at an arbitrary time $t > 0$ is a probabilistic replica of the process starting at $t = 0$; i.e., the time until the first arrival after t is a RV $\sim \exp(\lambda)$, and all subsequent interarrival times are independent of this first arrival epoch and of each other, and all these are distributed $\exp(\lambda)$

Proof:



Let Z be the distance from t until the first arrival after t..

Let's first condition on $N_t = 0$

$N_t = 0 \Rightarrow X_1 > t$ and $Z = X_1 - t$

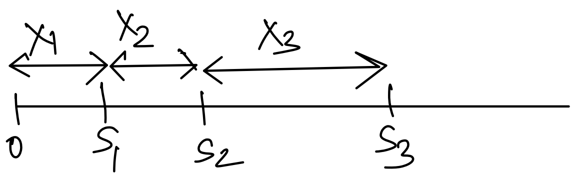
$$\Pr\{Z > z \mid N_t = 0\} = \Pr\{X_1 > z + t \mid N_t = 0\}$$
$$= \Pr\{X_1 > z + t \mid X_1 > t\}$$
$$= \Pr\{X_1 > z\} = e^{-\lambda z}$$

Prove for arbitrary $N_t = n$: Homework!

We have a Poisson process with rate λ

$\Rightarrow$ the inter-arrival times $X_1, X_2, \ldots \overset{iid}{\sim} \exp(\lambda)$

To prove: the counting process $N_t \sim Pos(\lambda t)$



1) Find joint density of $X_1, X_2, \ldots, X_n$.

Example: X_1, X_2, X_3

Joint pdf : $f_{X_1,X_2,X_3}(x_1,x_2,x_3) = \begin{cases} (\lambda e^{-\lambda x_1})(\lambda e^{-\lambda x_2})(\lambda e^{-\lambda x_3}) & \text{if } x_1>0, x_2>0, x_3>0 \\ 0 & \text{otherwise} \end{cases}$

$$\therefore f_{X_1,X_2,X_3}(x_1,x_2,x_3) = \begin{cases} \lambda^3 e^{-\lambda(x_1+x_2+x_3)} & , (x_1,x_2,x_3) \in (\mathbb{R}^+)^3 \\ 0 & , \text{otherwise} \end{cases}$$

Similarly

$$f_{X_1,X_2,\ldots,X_n}(x_1,\ldots,x_n) = \begin{cases} \lambda^n e^{-\lambda(x_1+x_2+\ldots+x_n)}, & (x_1,\ldots,x_n) \in (\mathbb{R}^+)^n \\ 0 & , \text{otherwise} \end{cases}$$

2) Find joint density of the arrival epochs $S_1, \ldots, S_n$ using the joint density of $X_1, \ldots, X_n$

$S_1 = X_1$

$S_2 = X_1 + X_2$

$S_3 = X_1 + X_2 + X_3$

$\vdots$

$S_n = X_1 + \ldots + X_n$

Suppose $n=3$

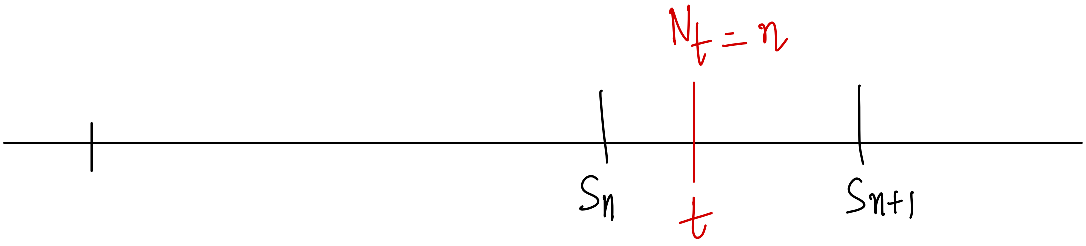
$$\left|\frac{\partial S}{\partial X}\right| = \begin{vmatrix} \frac{\partial S_1}{\partial X_1} & \frac{\partial S_1}{\partial X_2} & \frac{\partial S_1}{\partial X_3} \\ \frac{\partial S_2}{\partial X_1} & \frac{\partial S_2}{\partial X_2} & \frac{\partial S_2}{\partial X_3} \\ \frac{\partial S_3}{\partial X_1} & \frac{\partial S_3}{\partial X_2} & \frac{\partial S_3}{\partial X_3} \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = \underline{\underline{1}}$$

$$f_{S_1,S_2,S_3}(t_1,t_2,t_3) = \begin{cases} \lambda^3 e^{-\lambda t_3} & , 0<t_1<t_2<t_3 \\ 0 & , \text{otherwise} \end{cases}$$

Similarly, $f_{S_1,\ldots,S_n}(t_1,t_2,\ldots,t_n) = \begin{cases} \lambda^n e^{-\lambda t_n} & , 0<t_1<\ldots<t_n \\ 0 & , \text{otherwise} \end{cases}$

3) Required: find $P\{N_t = n\}$

$\{N_t = n\}$ is same as $\left\{(t_1, t_2, \ldots, t_n, t_{n+1}) \in (\mathbb{R}^+)^{n+1} \text{ s.t. } 0<t_1<t_2<\ldots<t_n \le t < t_{n+1}\right\}$



$$S_1 = t_1,\ S_2 = t_2, \ldots, S_n = t_n,\ S_{n+1} = t_{n+1}$$

$$A_{n,t} = \underbrace{\{(t_1, t_2, \ldots, t_n, t_{n+1}) \text{ s.t. } 0 < t_1 < t_2 < \ldots < t_n \le t < t_{n+1}\}}$$

$$= \{N_t = n\}$$

$$f_{S_1,\ldots,S_{n+1}}(t_1, t_2, \ldots, t_{n+1}) = \begin{cases} \lambda^{n+1} e^{-\lambda t_{n+1}}, & \underbrace{0 < t_1 < t_2 < \ldots < t_{n+1}} \\ 0, & \text{otherwise} \end{cases}$$

$$\underbrace{(t_1, t_2, \ldots, t_{n+1}) \in (\mathbb{R}^+)^{n+1}}_{0 < t_1 < t_2 \cdots < t_{n+1}}$$

$$\underbrace{f_{S_1,\ldots S_{n+1} | (N_t = n)}(t_1, t_2, \ldots, t_{n+1})} > 0$$

$$(t_1, t_2, \ldots, t_{n+1}) \in (\mathbb{R}^+)^{n+1}$$

$$\text{s.t. } 0 < t_1 < t_2 < \ldots < t_n \le t < t_{n+1}$$

$$\underbrace{f_{S_1,\ldots,S_{n+1} | N_t = n}(t_1, t_2, \ldots t_{n+1} \mid N_t = n)}_{\text{conditional pdf.}} = \begin{cases} \dfrac{\lambda^{n+1} e^{-\lambda t_{n+1}}}{P(N_t = n)} & \text{for all those } (t_1, t_2 \ldots t_n, t_{n+1}) \in A_{n,t} \\ 0, & \text{elsewhere} \end{cases}$$

$$f_{S_1,\ldots S_n | N_t = n} = \int_t^{\infty} f_{S_1,\ldots,S_{n+1} | N_t = n}\, dt_{n+1}$$

$$= \int_t^{\infty} \frac{\lambda^{n+1} e^{-\lambda t_{n+1}}}{P(N_t = n)}\, dt_{n+1}$$

$$= \frac{1}{\mathbb{P}(N_t=n)} \lambda^n \underbrace{\int_t^\infty \lambda e^{-\lambda t_{n+1}} \, dt_{n+1}}$$

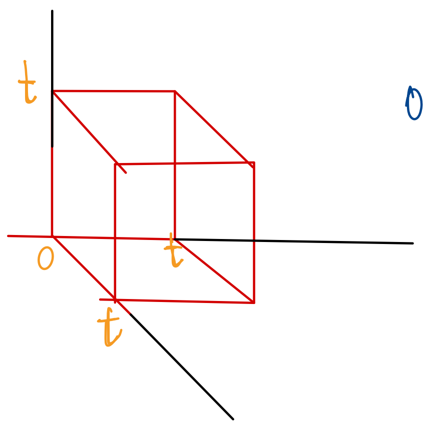
$$f_{S_1,\dots,S_n \mid N_t=n} = \frac{1}{\mathbb{P}(N_t=n)} \lambda^n . e^{-\lambda t}$$

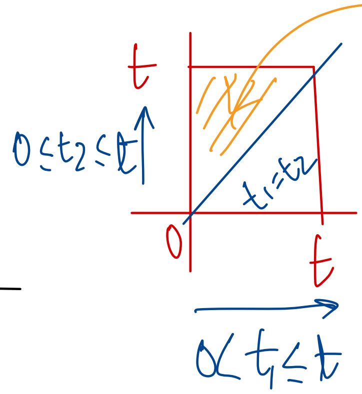
$$f_{S_1,\dots S_n \mid N_t=n}\left(S_1=t_1, S_2=t_2, \dots, S_n=t_n\right) = \underbrace{\frac{1}{\mathbb{P}(N_t=n)} \cdot \lambda^n e^{-\lambda t}}$$

for all $(t_1,\dots,t_n) \in (\mathbb{R}^+)^n$

s.t $0<t_1<t_2 \dots < t_n \le t$

$$f_{S_1,S_2,S_3 \mid N_t=3}\left(S_1=t_1, S_2=t_2, S_3=t_3\right) = \begin{cases} \dfrac{1}{\mathbb{P}(N_t=3)} \lambda^3 . e^{-\lambda t}, & \underline{0<t_1<t_2<t_3\le t} \\ 0 & \text{otherwise} \end{cases}$$





Sequences of different $0<t_1<t_2\le t$

All such (t_1,t_2) have conditional density $= \lambda^2 . e^{-\lambda t}$ (conditioned on $N_t=2$).

The conditional density is constant on the set $\overbrace{\{(t_1,\dots,t_n) \in (\mathbb{R}^+)^n : 0<t_1<t_2<\dots<t_n\le t\}}^{B_n^t}$

and the constant $= \dfrac{1}{\underbrace{\text{n-D volume of the set } B_n^t}_{= \frac{t^n}{n!}}}$

$$\therefore \frac{\lambda^n . e^{-\lambda t}}{P\{N_t=n\}} = \frac{n!}{t^n} \Rightarrow P(\{N_t=n\}) = \frac{(\lambda t)^n . e^{-\lambda t}}{n!}$$

ie $N_t \sim Pos(\lambda t)$.

Volume:

$$\underbrace{\{(t_1,\dots,t_n)\in(\mathbb{R}^+)^n \text{ s.t } 0<t_1<t_2<\dots<t_n\le t\}}: B_n^t$$

$$= \frac{t^n}{n!}$$

Why?

$$\underbrace{f_{S_1,\dots S_n \mid N_t=n}} = \frac{1}{\text{volume of this set}} = \frac{n!}{t^n}$$

$$\frac{\lambda^n e^{-\lambda t}}{P\{N_t=n\}} = \frac{n!}{t^n} \Rightarrow P\{N_t=n\} = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$$

$$= Pos(\lambda t)$$

$$\boxed{N_t \sim Pos(\lambda t)}$$

Each of the sequences $(t_1,\dots,t_n)$ satisfying $0<t_1<\dots<t_n\le t$ have the same cond. density $\underbrace{e^{-\lambda t}\cdot\lambda^n}$ (independent of $t_1,\dots,t_n$; only depends on t) and the cumulative prob of all such sequences is $\frac{t^n}{n!}$ $\therefore$ $f_{S_1,\dots S_n}(t_1,\dots t_n \mid N_t=n)$ for each such t is $\frac{1}{\text{volume}} = \frac{n!}{t^n}$.

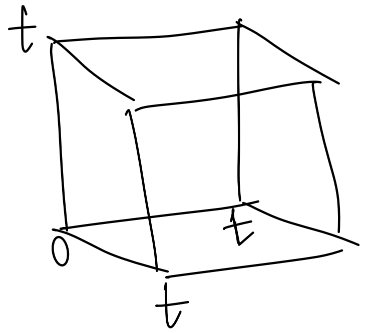
$$\underbrace{\int_{B_n^t} \underbrace{f_{S_1,\dots,S_n}(t_1,\dots t_n \mid N_t=n)}_{\text{constant } e^{-\lambda t}\cdot\lambda^n} dt_1\dots dt_n}_{\text{total prob. of all seq. satisfying } 0<t_1<\dots<t_n\le t} = \text{volume}(B_n) = \frac{t^n}{n!}.$$

(For instance: if X distributed in $(0,a)$ with each x_1 has the same const. $f_{X_1}(x)$ ind. of x, then $f_{X_1}(x) = 1/a$. Similarly if X_1,X_2 distributed in the area A, with each $(X_1=x_1, X_2=x_2)$ have $f_{X_1,X_2}(x_1,x_2)=c$ indep. of x_1,x_2 then $\int f_{X_1,X_2}(x_1,x_2)\,dx_1 x_2 = A$ (the Area occupied) and hence $f_{X_1,X_2}(x_1,x_2) = \frac{1}{A}$. for all possible (x_1,x_2) lying in A.)

Why $\frac{1}{n!}$ of the total volume t^n?

Illustration with $n = 3$.

$(\mathbb{R}^+)^3$ $\quad$ $(t_1, t_2, t_3) \in (\mathbb{R}^+)^3$ s.t $0 \le t_1 \le t$, $0 \le t_2 \le t$, $0 \le t_3 \le t$



(t_1, t_2, t_3) $\quad$ $t_1 < t_2 < t_3$

If $t = 2$

$\underbrace{(1, 1.25, 0.5)}$ $\quad$ $t_1 < t_2 < t_3$

(t_1, t_2, t_3)

$\boxed{t_1 < t_2 < t_3}$

$t_1 < t_3 < t_2$

$t_2 < t_1 < t_3$

$t_2 < t_3 < t_1$

$t_3 < t_1 < t_2$

$t_3 < t_2 < t_1$

$\frac{t^3}{3!}$